Fiscal Year:	FY 2014	**Task Last Updated:**	FY 09/09/2014
PI Name:	Tash, Joseph S Ph.D.		
Project Title:	Long Term Space Flight Impacts on Male Reproductive Health		
Division Name:	Space Biology		
Program/Discipline:	SPACE BIOLOGY		
Program/Discipline--Element/Subdiscipline:	SPACE BIOLOGY--Developmental biology		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Cell & Molecular Biology (2) Animal Biology: Vertebrate		
Space Biology Cross-Element Discipline:	(1) Reproductive Biology (2) Developmental Biology		
Space Biology Special Category:	(1) Translational (Countermeasure) Potential		
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City:	Kansas City	**State:**	KS
Zip Code:	66160-0001	**Congressional District:**	3
Comments:			
Project Type:	Flight	**Solicitation / Funding Source:**	2007 Space Flight: Bion-M1 NNH07ZTT001N
Start Date:	08/01/2009	**End Date:**	07/31/2015
No. of Post Docs:	2	**No. of PhD Degrees:**	3
No. of PhD Candidates:		**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	2
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA ARC
Contact Monitor:	Smith, Jeffrey	**Contact Phone:**	650-604-0880
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Flight Program:	Bion/Shuttle		
Flight Assignment:	STS-131 ; STS-133 ; STS-135 ; Bion M1 NOTE: Extended to 7/31/2015 per A. Chu/ARC and NSSC information (Ed., 8/5/14) NOTE: Extended to 7/31/2014 per A. Chu/ARC (Ed., 5/7/13)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Roby, Katherine (University of Kansas Medical Center) Gupta, Vijayalaxmi (University of Kansas Medical Center) Holets, Lesya (University of Kansas Medical Center)		
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Task Description:	Published data demonstrates that short term microgravity (μG) (~2 wks) produces small but significant reductions in spermatogenesis in rats. However, long term ground-based μG models using the rodent hindlimb unloading (HLS) (> 3 wks), with surgical prevention of cryptorchidism, show dramatic testicular atrophy and loss of spermatogenesis leading to sterility. These results and related literature in humans suggest that longer duration space flight may have negative impacts on spermatogenesis and compromise male health and increase cancer risk. Thus, the testable hypothesis driving this proposal is: Exposure to μG 3 wks or longer causes disruption of spermatogenesis in mice. This proposal addresses the Bioastronautic Roadmap (BR) Risk 30: Chronic and Degenerative Tissue Risks in that long term μG-induced testicular atrophy will increase the risk for long term health problems post-flight including increased cancer risk and sterility. The hypothesis will be tested with the following specific aims: 1. Identify specific histopathology and dysfunction in spermatogenesis and testis function in mice exposed to LEO for 30d (as determined by mission duration) and compare to ground-based controls. 2. Quantify declines in sperm motility and morphology in cauda epididymal sperm from male mice exposed to LEO for 30d (as determined by mission duration) and compare to ground-based controls. 3. Determine dysregulation in gene transcription of markers of spermatogenic regulation and inflammatory markers in testes from mice exposed to LEO for 30d (as determined by mission duration) and compare to ground-based controls. 4. Determine declines in fertility and fecundity of male mice exposed to μG (This aim will depend upon availability of flight animals for mating trials). The opportunity to expose male mice for 30d of μG in the new BION flights represents the first to directly evaluate this reproductive health problem directly in long term μG under properly controlled conditions. These experiments will validate the ground-based long term HLS rodent model and provide proof that male reproductive health is put at risk by extended exposure to μG. The experiments will provide an initial data set for development of appropriate countermeasures to the negative impact of long term space flight on endocrine and reproductive physiology compromising male health.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Female reproductive studies: The observed down-regulation of estrogen receptor ERa, combined with the histological changes, suggests that the animals stopped cycling in space flight and estrogen support was severely impaired. The observations are also significant in that the results were verified on three separate Space Shuttle experiments (STS-131, STS-133, and STS-135). If ERa is also down regulated in other tissues, this would indicate a common mechanism underlying space-flight induced degradation of other estrogen dependent systems including bone loss, muscle weakness, immune response, and wound repair. A broad effect of space flight on ERa receptor levels and/or signaling will offer a common pathway to generate countermeasures for space flight. Since similar changes occur in the same organ systems during the aging process, the countermeasures would have application for improving quality of life in the same organ systems during the aging process. Male reproductive health: The experiments outlined in this proposal will be the first to determine directly, under controlled space flight conditions, whether exposure to long microgravity (μG) disrupts sperm production and compromises male reproductive health in mammals. The unique opportunities offered in the new BION 30d+ rodent flights are the first to extend space flight beyond all previous μG exposures for rodent models on any flight platform. As such, the planned experiments will be the first to directly assess a long term health risk that has been identified by ground-based space flight animal and human models. Thus the hypothesis driving the research: that long term space flight has negative impacts on male reproductive health. Specific Aim 1. Identify specific histopathology and dysfunction in spermatogenesis and testis function in mice exposed to LEO for 30d (as determined by mission duration) and compare to ground-based controls. Specific Aim 2. Quantify declines in sperm motility and morphology in cauda epididymal sperm from male mice exposed to LEO for 30d (as determined by mission duration) and compare to ground-based controls. Specific Aim 3. Determine dysregulation in gene transcription of markers of spermatogenic regulation and inflammatory markers in testes from mice exposed to LEO for 30d (as determined by mission duration) and compare to ground-based controls. Space flight impacts on reproductive health include both microgravity and radiation environmental factors. Space flight changes are an excellent accelerated analogue to the aging process on Earth. The 30 day BION M1 mission will combine the longest exposure and the largest number of mice in a single mission and represent a significant duration relative to the complete spermatogenesis cycle in the testis. Changes in sperm motility, spermatogenesis, gene transcription, and DNA damage in long term space flight will offer the first comparison for identical male reproductive health outcome measures for the normal aging process on Earth.
	The BION M1 flight was the first opportunity to directly test the hypothesis that prolonged space flight negatively affects testicular function and spermatogenesis. Seven mature male mice were flown in orbit for 30d. Asynchronous diet and age-matched ground control (GC) mice were maintained in the same BION hardware on the ground for 30d. For histology, testis and epididymis were fixed in Bouin's solution, washed in 70% ETOH, paraffin-embedded and processed for hematoxylin and eosin staining. For RNA and protein extraction tissues were harvested in RNAlater and stored at -80°C. RNA was subjected to real-time RT PCR. Next Generation sequencing was used to compare gene expression profiles between flight and GC mouse testes. We report preliminary results from analysis of a subset of flight and GC animals for histological modifications, and testicular/spermatogenesis gene expression changes. Remarkably, only 43 genes showed significant changes (q = .05) of 14,360 expressed genes in flight, and 14,374 genes in GC (FPKM>1). In the flight group, elevated expression of testicular and epididymis-specific genes included LCN8 (lipocalin 8) and Defb41, involved in testicular exocrine secretion; Cst11 (cystatin11), responded for gonadal development; Cldn2 (claudin 2) relevant to the blood-testes and blood-epididymal barrier; RNase10, required for post-testicular sperm maturation and male fertility. Significantly down-regulated genes included FABP4- fatty acid binding protein 4, androgen-dependent CFD-complement factor D, Car3, carbonic anhydrase 3, inflammatory marker IL-12. Quantitative RT PCR confirmed dramatic down-regulation of FABP4 (over 20 fold), CFD (5- fold) and Car3 (2-fold) in the flight group testis. Testis estrogen receptor-alpha expression was also reduced by μG exposure ($P<0.05$).

Task Progress:	We confirmed a 4-fold increased Defb41 in flight vs GC. Differentially expressed genes involved in several canonical pathways included immune responses, metabolic, and endocrine functions. Preliminary assessment of the testis histopathology suggests some changes in integrity /organization of the seminal epithelium in the space flight testes. Further histological and molecular analysis will help us to understand the effects of gravity for male reproductive system. The mice were sexually mature at launch, so that any observed changes can be attributed to space-flight induced changes in mature testis function and not alterations in a) ongoing maturation of post-natal or b) pre-pubertal testis development, or c) continuing sexual maturation. As such, these animals would be a rodent analogue to adult male astronauts. The RNA quality used from the replicate mice testes from all groups was in excellent condition with no evidence of degradation. The effect of space flight was limited to an extremely small subset of the entire expressed genome with, using high stringency, significant changes in gene expression of only 43 genes out of ~14,370 detected genes. This represents a remarkable 0.3% of the entire genome. This means that the space flight effect is very focused to a small subset of genes in the testis. The strength in this finding is that any pathways identified have a high likelihood of being directly related to space flight effects. This is a valid conclusion since the animals in both flight and BION hardware ground control groups were no different, and unremarkable with respect to body weight (nutrition) and health post-flight. This conclusion is supported by the fact that they were also not significantly different by the same parameters from the parallel vivarium controls that were used as an internal reference standards since the flight and ground control BION groups were asynchronous.
Bibliography Type:	Description: (Last Updated: 01/24/2022)
Abstracts for Journals and Proceedings	Holets-Bondar LM, Gunewardena S, Knapp J, Tash JS. "BION-M1: Space flight for 30d alters the expression of testis- and epididymis-specific genes in male mice." To be presented at the 30th Annual Meeting of the American Society for Gravitational and Space Research, Pasadena, CA, October 22-26, 2014. 30th Annual Meeting of the American Society for Gravitational and Space Research, Pasadena, CA, October 22-26, 2014. Program and abstracts, in press, as of September 2014. , Sep-2014
Articles in Peer-reviewed Journals	Ronca AE, Baker ES, Bavendam TG, Beck KD, Miller VM, Tash JS, Jenkins M. "Effects of Sex and Gender on Adaptation to Space: Reproductive Considerations." Journal of Women's Health. 2014:In press. , Sep-2014